

A Computational Approach for the Prediction of NMR Spin-Spin Coupling Constants in Deep Eutectic Solvents

V. Castor^a, T. Potez^a, M. Hédouin^a, F.M. Perna^b, J. Maddaluno^c, M. Durandetti^a, V. Capriati^b, V. Tognetti^a and H. Oulyadi^a

^aUniv Rouen Normandie, INSA Rouen Normandie, CNRS, Normandie Univ, Institut CARMen UMR 6064, INC3M FR 3038, F-76000, Ruen France; ^bDipartimento di Farmacia-Scienze del Farmaco, Università degli Studi di Bari Aldo Moro, Consorzio CINMPIS, via E. Orabana 4, 70125 Bari, Italy; ^cLaboratoire LAMS, Sorbonne Université, CNRS, UMR 8220, 4 place Jussieu, 75252 Paris cedex 05, France; e-mail victoria.castor-villegas@univ-rouen.fr

00 Introduction

Reline, a 1:2 mixture of choline chloride and urea, is a widely studied deep eutectic solvent. Its properties depend strongly on its hydrogen bond network, yet its NMR coupling remains insufficiently characterised.¹

01 The approach

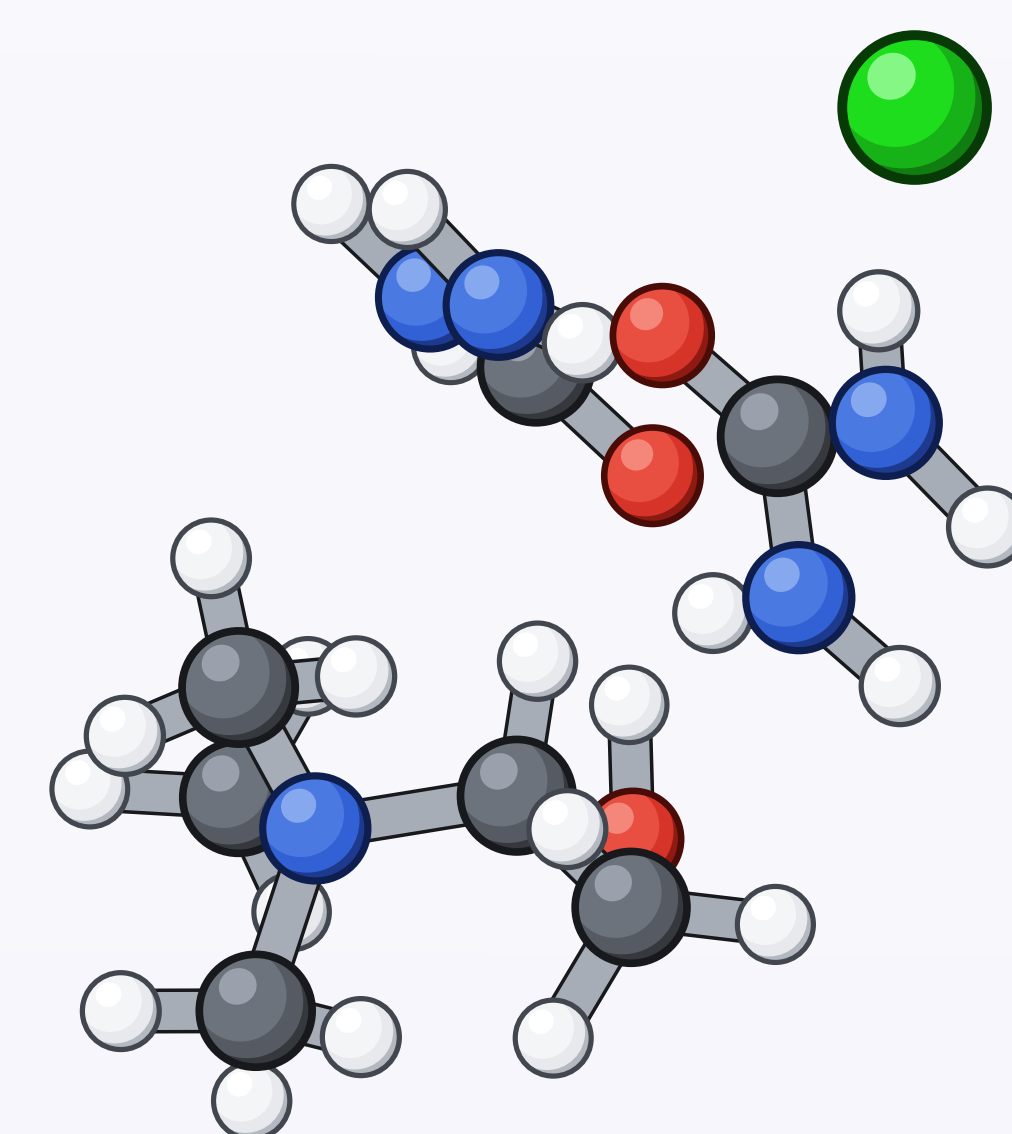
An automated first-principles protocol predicts spin-spin coupling constants in reline under realistic conditions. The workflow is performed entirely within the Amsterdam Modelling Site,² combining molecular dynamics, DFT calculations at PBE/TZ2P and PBE/TZ2P-J levels,³ coupling constants under Ramsey's formalism⁴ and QTAIM⁵ & CDFT⁶ descriptors.

02 The system

Urea
C(=O)(N)N

Choline
C[N+](C)(C)CCO

Chloride
[Cl-]



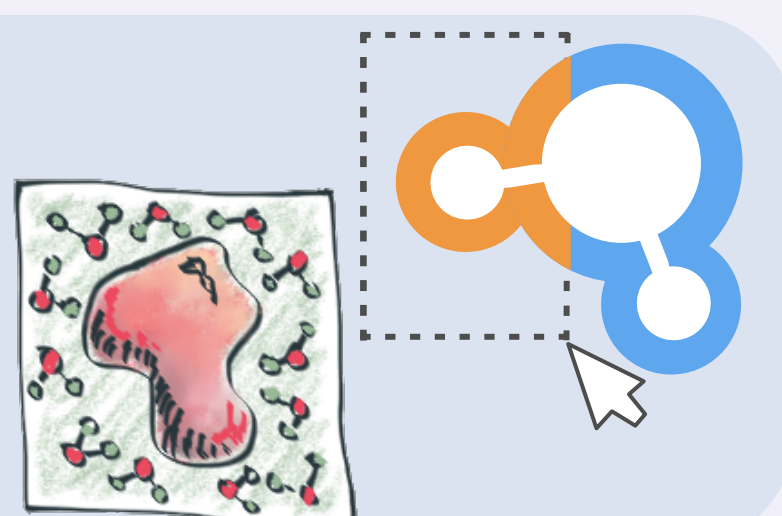
@ 303.15 K & 0.1 MPa; 1.1945 gcm⁻³

03 Automated protocol *a pipeline, end to end*

INPUT

Build systems from SMILES

packmol by AMS GUI



SAMPLING

Molecular Dynamics (GAFF)

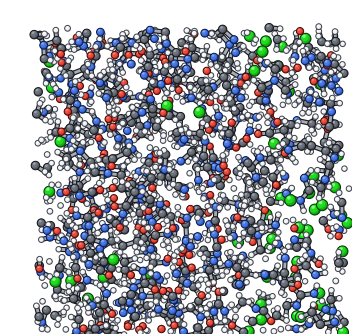
Thermostat: Equilibration Berendsen, Production NHC



PLAMS

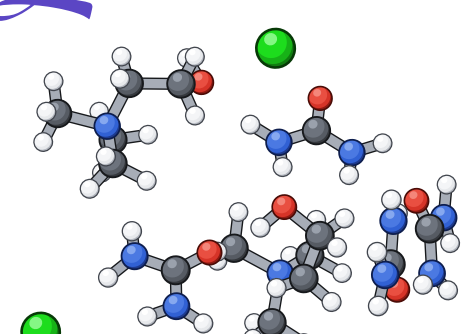
Python Library for Automating Molecular Simulation

1 Read rkf step



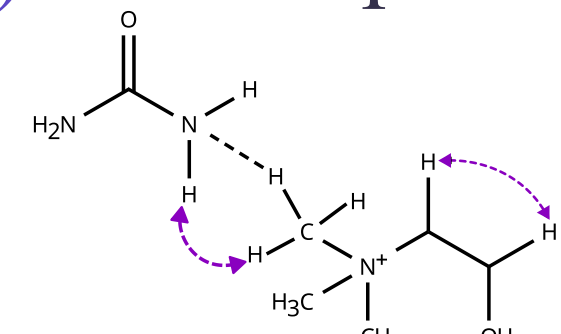
save the coordinates every 200 fs

2 Carve cluster



all molecules in 10 Å with the ratio (1:2)

3 Select pairs



4 Single Point

ADF (PBE/TZ2P)



5 Database



Analysis:

Distances

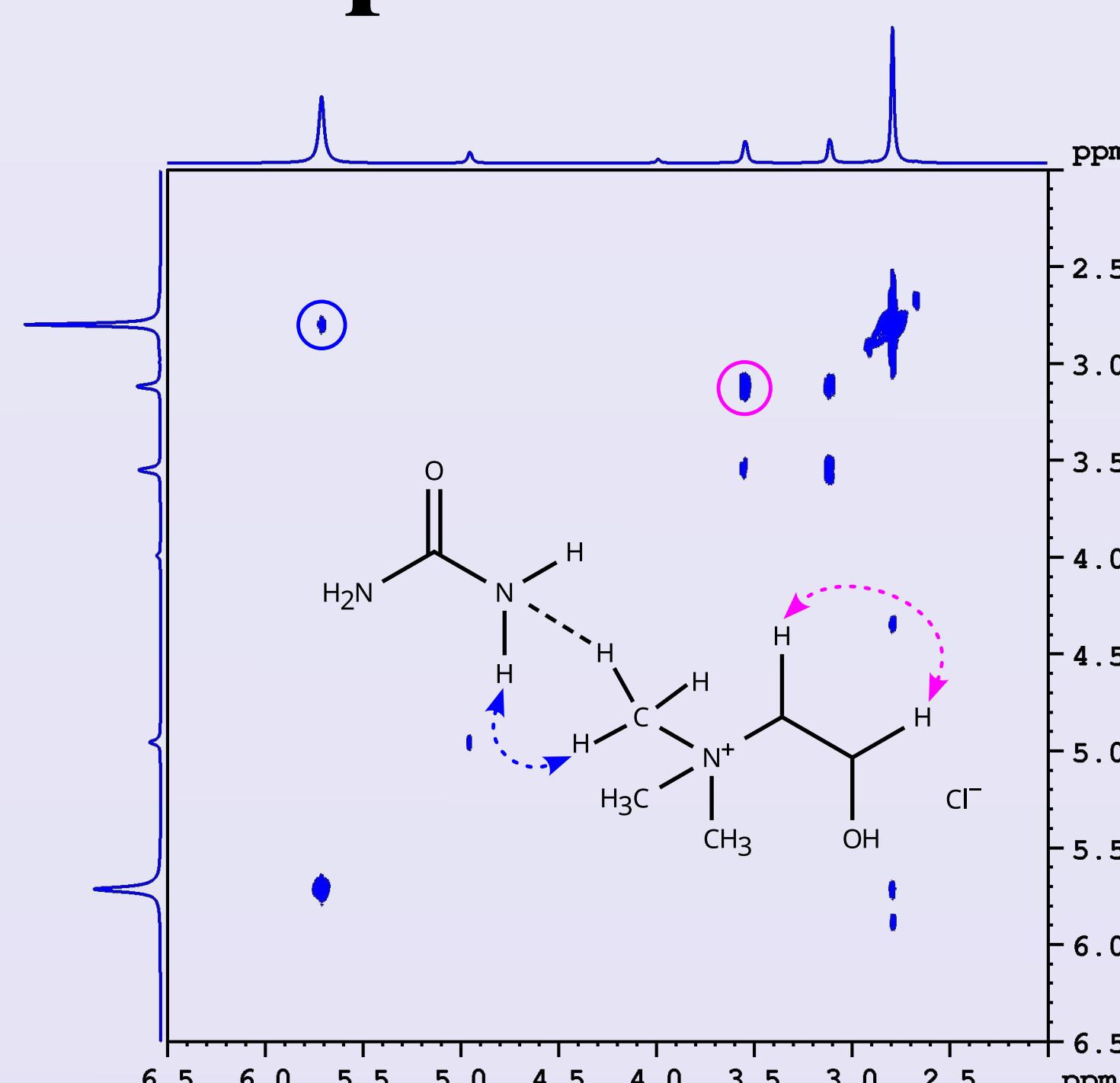
J

Conceptual DFT

QTAIM



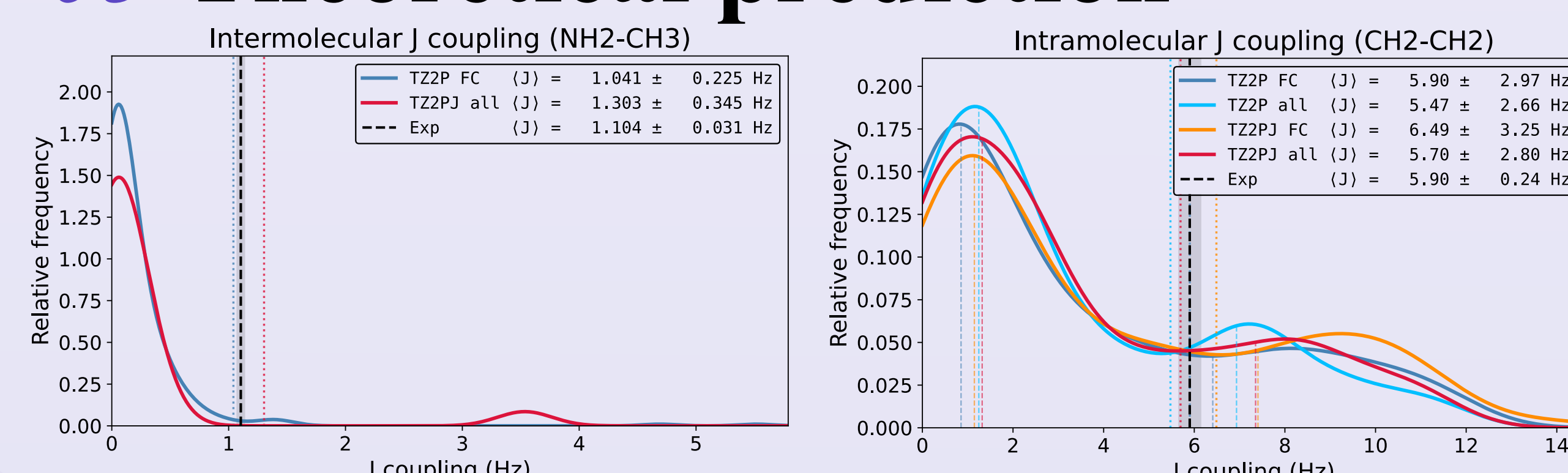
04 Experimental data



$J_{HH} = 1.104 \pm 0.031$ Hz
 $J_{HH} = 5.90 \pm 0.24$ Hz

¹H CT-COSY spectrum of Reline
(0.2 % w/w H₂O)
($\Delta = 40$ ms; 303 K; 600 MHz)

05 Theoretical prediction



06 Conclusions

The protocol connects MD sampling with representative clusters, extending spin-spin coupling predictions towards a realistic description of reline.

Acknowledgments

Université de Rouen —CARMen Institute— and CNRS support this research. Thanks, CRIANN (Centre Régional Informatique et d'Applications Numériques de Normandie) for the computational time.

References

- [1] I. Delso, C. Lafuente, J. Muñoz-Embid, M. Artal, J. Mol. Liq., 290, 111236 (2019).
- [2] E. J. Baerends, N. F. Aguirre, N. D. Austin, J. Autschbach and A. Yakovlev, J. Chem. Phys. 162, 162501 (2025).
- [3] J. Autschbach, ChemPhysChem 10, 2274-2283 (2009).
- [4] N. F. Ramsey, Phys. Rev. 91, 303-307 (1953).
- [5] R. F. W. Bader. Oxford University Press, (1994).
- [6] R. G. Parr and W. Yang, Annu. Rev. Phys. Chem. 46, 701-728 (1995).